

37495 Statistical Design and Models for Evaluation Studies

Computer Lab 3

Question 1

A manufacturer is investigating the tensile strength of a new synthetic fibre. The material will be used to make shirts. Strength is believed to be a function of the percentage of cotton in the fibre. The manufacturer chooses 5 proportions of cotton and measures the strength of 5 pieces of fibre. The data from the experiment appear below (higher values of *strength* correspond with higher cotton strength).

Table 1: Tensile strength for varying cotton percentages

cotton	1	2	3	4	5
15%	7	7	15	11	9
20%	12	17	12	9	18
25%	14	10	19	18	18
30%	19	25	22	19	23
35%	7	10	19	15	11

- Write down the linear model for this study and explain the components in the model.
- State the assumptions necessary for an analysis of variance to be valid.
- Compute the analysis of variance table.
- Produce the appropriate residual plots.
- Write a 3-5 line summary of the results.
- Perform an appropriate non-parametric test. Do the results agree with the corresponding parametric analysis?

Question 2

A temperature accelerated life test was performed on a type of sheathed tubular heater. Six heaters were tested at each of four temperatures 825°C, 885°C, 905°C, 930°C. The number of hours to failure was recorded for each of the 24 heaters in the study.

Table 2: Hours to failure at different temperatures

temperature	1	2	3	4	5	6
825	1953	2135	2471	4727	6134	5314
885	1190	1286	1550	2125	2557	2845
905	651	837	848	1038	1141	1543
930	511	651	651	652	688	480

- By performing appropriate hypothesis tests on the residuals of a fitted model, determine whether the assumptions necessary for an analysis of variance hold.
- By performing appropriate hypothesis tests on the residuals of a fitted model, determine whether the assumptions are satisfied if the reciprocal of the original response ($1/\text{hours}$ to failure) is used instead.
- Use a Box-Cox transformation to determine the most appropriate transformation of the response.
- Using the model in c). test whether the time to failure is related to temperature.
- Perform an appropriate non-parametric test. Do the results agree with the corresponding parametric analysis?

Question 3

An experiment was run to compare the crank rates needed to keep a bicycle at a particular speed when the bicycle was on flat ground and in 12th gear. The speeds chosen were 8, 16, 24, 32, and 40 km/h. The results are given in the table below.

Table 3: Crank rates to keep bicycles at different speeds

speed	1	2	3	4
8 km/h	10	15	25	18
16 km/h	32	37	22	24
24 km/h	42	47	45	42
32 km/h	59	61	61	60
40 km/h	75	74	75	75

- By performing appropriate hypothesis tests on the residuals of a fitted model, determine whether the assumptions necessary for an analysis of variance hold when an untransformed response is used.
- Use a Box-Cox transformation to determine the most appropriate transformation of the response.
- Using the most appropriate model, test whether there is a significant difference in the crank rate to obtain different speeds.
- Perform an appropriate non-parametric test. Do the results agree with the corresponding parametric analysis?